From Firms to Transaction-Scoped Institutions: A Coasean Re-Examination Under Costless Bilateral Enforcement

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Abstract

Coase [1937] argued that firms exist because the cost of discovering prices, negotiating terms, and enforcing contracts in open markets sometimes exceeds the cost of hierarchical management. Williamson [1985] refined this into Transaction Cost Economics (TCE), identifying asset specificity, uncertainty, and frequency as drivers of vertical integration. Both arguments rest on an implicit assumption: that contract enforcement between strangers is costly, slow, and jurisdiction-dependent. This paper examines what happens when that assumption no longer holds.

We take as given the existence of a settlement primitive—developed formally in the companion mechanism paper, with a verified Solidity implementation in the companion implementation paper—that prices the cost of trust at exactly 2× the transaction value in temporarily locked capital, with no recurring extraction, no arbitrator, and no jurisdictional dependency. Treating this primitive as exogenous, we ask three questions. First, what does the existence of such a primitive imply for the Coasean threshold that defines the boundary of the firm? Second, which functions of the firm dissolve under it, and which persist in revised form? Third, why is the implication structurally invisible—and what does that invisibility itself say about the architecture of contemporary coordination infrastructure?

Our argument is that the firm bundles several functions historically inseparable in practice but logically distinct: coordination of production, capital aggregation, organizational knowledge, risk pooling, and legal personhood. A costless bilateral enforcement primitive dissolves the first while leaving the others largely intact. The result is not the abolition of durable institutions but a structural shift: the *coordination firm*, whose value is internal management of a bounded labor pool, loses its economic rationale where the primitive is applicable; the *capital firm*, whose value is durable asset ownership and risk-bearing, persists in revised wrappers. We close with a Gramscian observation about why this shift is harder to see than to specify.

Keywords: transaction cost economics, theory of the firm, smart contracts, coordination economics, post-firm organization, cultural hegemony

1 Introduction

The boundary of the firm has been the central object of theoretical attention in organizational economics since Coase [1937]. Williamson's Transaction Cost Economics (TCE) [Williamson, 1985] identified asset specificity, uncertainty, and frequency as the conditions under which hierarchical governance dominates market exchange. The property-rights theory of the firm [Hart, 1995] located the firm boundary at the allocation of residual control rights over non-contractible decisions. Across these traditions, a common premise is that market contracting between

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strangers is expensive: search is costly, negotiation is costly, and enforcement—through courts, arbitrators, or platforms—is costly, slow, and jurisdiction-dependent.

This paper asks what happens to that boundary when one of the underlying costs—contract enforcement between mutually distrusting parties—collapses to a fixed, jurisdiction-independent capital lockup. The collapse is not a thought experiment: the companion mechanism paper presents the mechanism with formal proofs of cooperation as the unique surviving Nash equilibrium, and the companion implementation paper presents a verified Solidity implementation audited through TLA^+ model checking, property-based fuzzing, symbolic execution, and specification checking.

We treat that primitive as exogenous. The contribution of this paper is institutional-economic: given the primitive, what follows for the theory of the firm? We argue three things. First, the Coasean threshold shifts inward—narrowly, in a precise sense we make explicit. Second, the firm's various functions dissolve at different rates: coordination dissolves where the primitive is applicable; capital aggregation, organizational knowledge, risk pooling, and legal personhood persist. Third, the practical difficulty of communicating this shift is itself diagnostic: the institutions whose necessity becomes contingent are precisely the institutions whose necessity has achieved hegemonic invisibility.

Paper organization. Section 2 summarizes the settlement primitive, citing the companion paper for the formal development. Section 3 develops the Coasean threshold shift. Section 4 distinguishes the coordination firm from the capital firm and locates the dissolution. Section 5 examines token denomination as a coordination signal that replaces standing platform credentials. Section 6 offers a Gramscian reading of why the shift is hard to see. Section 7 concludes.

2 The Settlement Primitive

We summarize here only the properties of the primitive that the present argument uses. The full definitions and theorems are in the mechanism paper; the implementation and verification in the implementation paper.

Asymmetric bonding. For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. Only the buyer can trigger resolution. On resolution, the buyer recovers P (net cost: $-P$) and the seller recovers $2G + P$ (net gain: $+P$). On non-resolution, both bonds remain locked indefinitely.

Equilibrium properties. In the resulting one-shot bonding game, cooperation weakly dominates defection for all parties, and the cooperative profile is the unique outcome surviving iterated elimination of weakly dominated strategies. The result extends to N-party process trees through *progressive collateralization*, with the seller's risk-to-reward ratio growing with chain depth.

Atomic resolution and peer enforcement. All orders within a process resolve atomically—all or nothing—which induces a weakest-link subgame among sellers. This produces an endogenous coordination pressure analogous to peer enforcement in joint-liability microfinance, but obtained without repeated interaction, local information, or social punishment technology.

No escape hatches. The primitive has no timeout, no admin override, no governance mechanism, no protocol fee, and no upgrade path. The companion paper proves that any such escape hatch weakens the equilibrium.

What the primitive does *not* provide. For the institutional argument that follows, two non-properties matter as much as the properties. The primitive does not provide capital aggregation across projects, does not provide pooled risk-bearing across uncorrelated ventures, and does not provide legal personhood for off-chain liability. We return to these absences in Section 4.

3 The Coasean Threshold Shift

Coase's central insight is that firms exist because market transaction costs—searching for partners, negotiating terms, enforcing agreements—sometimes exceed the cost of hierarchical management. Williamson refined this into TCE, identifying three conditions under which hierarchical governance dominates market exchange: high asset specificity, high uncertainty, and high frequency.

The settlement primitive changes the calculus by collapsing one of these costs to a fixed, predictable capital lockup. Consider the three transaction costs separately.

Search costs. Public on-chain signals (settlement history, accepted-token lists, attested capabilities) allow autonomous discovery through shared coordination graphs. For agents able to read blockchain state, search cost approaches zero. The substantive change is not that search is cheaper—platforms already make search cheap—but that the search index is non-proprietary: no platform owns the discovery function, and the data is not extracted as rent.

Negotiation costs. Off-chain commitment building (EIP-712 typed structured data) lets parties iterate on terms without gas costs. The negotiation is bilateral and direct; counterparties can discover and iterate through shared graph visibility rather than through a standing intermediary's matching function. The substantive change here is asymmetric: the *cost* of negotiating is roughly unchanged from web2 baselines, but the *counterparty space* from which one can negotiate is no longer mediated by a platform's terms of service.

Enforcement costs. This is the decisive variable. In the pre-primitive world, enforcement requires courts (slow, expensive, jurisdiction-dependent), arbitrators (trusted third parties), or standing intermediaries that internalize coordination and charge for it. The primitive prices enforcement at exactly $2\times$ the transaction value—a fixed, predictable, jurisdiction-independent cost requiring no third party.

Proposition 3.1 (Coasean Threshold Shift)**.** *For any transaction where $2P$ (the capital lockup cost) is less than the firm's coordination overhead for the same transaction, the rational organizational choice is market exchange via the primitive rather than hierarchical integration.*

Theorem 3.1 is a comparative-statics statement, not a sweeping institutional prediction. It identifies a class of transactions for which the make-or-buy boundary tilts toward *buy*—specifically, transactions where the time-weighted opportunity cost of locked capital is lower than the standing overhead of internalizing the coordination function. The class is non-empty (most retail-scale transactions in mature financial environments satisfy it) and bounded (very-high-frequency, very-low-margin, or very-long-duration transactions do not). The proposition does not claim that all transactions migrate; it claims that the migration threshold is now defined by capital opportunity cost rather than by the cost of standing institutional infrastructure.

Engagement with TCE. Williamson's three drivers of vertical integration deserve direct treatment. *Asset specificity* continues to favor hierarchy: a specialized asset whose value is contingent on a specific counterparty cannot be efficiently coordinated through a sequence of short-lived bonded transactions, since the holdup risk is in the asset's pre-transaction sunk investment, not in the transaction itself. *Uncertainty* cuts both ways: contract incompleteness favors hierarchy, but the primitive's evidence trail (timestamped commitments, signed disclosures) reduces some classes of uncertainty—specifically those that hierarchical authority historically resolved by managerial fiat. *Frequency* is the condition under which the shift is most pronounced: high-frequency exchanges with low asset specificity are exactly where the standing firm's overhead is hardest to amortize and the primitive's per-transaction capital lockup is easiest to absorb.

The proposition therefore narrows rather than overturns TCE. It identifies a region of the asset-specificity / uncertainty / frequency space where hierarchical governance was previously the equilibrium outcome but is no longer.

4 From Coordination Firm to Capital Firm

The implication of Theorem 3.1 is structural: the boundary of the firm shifts inward in a precise region. The stronger implication is organizational rather than merely contractual. When enforcement is priced directly in bonded capital, the standing firm ceases to be the default container of production for the class of transactions identified above. What emerges in that region is a *transaction-scoped institution*: a temporary assembly of directly bonded counterparties that forms around a process, coordinates for the duration of that process, and dissolves at settlement.

The dissolution is narrower than Theorem 3.1 alone may suggest. The firm historically bundles at least five distinct functions:

(i) **Coordination of production** — aligning the activities of multiple value-adders to produce a finished output.

(ii) **Capital aggregation** — pooling capital across projects to fund investment that no single participant can finance alone.

(iii) **Organizational knowledge** — the accumulated tacit procedural and relational knowledge that is hard to codify.

(iv) **Risk pooling** — diversifying across uncorrelated ventures to smooth participant income.

(v) **Legal personhood** — providing a contractable entity for liability, regulatory compliance, and external counterparty relationships.

The settlement primitive addresses (i) directly. It does not address (ii)–(v). The remaining functions persist as requirements that some durable institution must satisfy; they likely reconstitute around token-denominated treasury communities whose members repeatedly participate in compatible process trees—structures that resemble firms at the meta-layer.

The precise claim is therefore: the *coordination firm*, whose value is internal management of a bounded labor pool, loses its economic rationale where the primitive is applicable; the *capital firm*, whose value is durable asset ownership and risk-bearing, persists in revised wrappers.

Key components and individually addressable capability. Each seller node in a process tree corresponds to what we term a *key component*—the irreducible asset or capability that provides competitive advantage in a specific market. In the firm model, key components are embedded inside organizational hierarchies: the cook's skill, the kitchen's equipment, the courier's vehicle are all subsumed under the firm's legal identity. In the primitive model, each key component is represented directly by a seller wallet address. The bonded process tree does not merely flatten the firm—it makes each value-adding capability individually addressable, individually compensated, and individually accountable. The productive asset *is* the identity.

Nothing in the primitive requires that a node be a natural person or a corporation. Any actor capable of controlling an address and bearing capital risk—a human, a software agent, a machine operator, or a treasury—can occupy a node in the process tree. The accountable unit is the bonded counterparty, not the firm.

What this means for the property-rights theory. The property-rights tradition [Hart, 1995] locates the firm boundary at the allocation of residual control rights over non-contractible decisions. Under the primitive, certain previously non-contractible decisions become contractible: schema-typed attestations and atomic-resolution conditions encode behaviors that historically required human discretion (acceptance, dispute, quality verification). The class of residual non-contractible decisions therefore shrinks. The property-rights logic still applies to what remains; the boundary moves because the set of contractible behaviors expands.

5 Token Denomination as Coordination Signal

In the primitive, the choice of denomination token is itself a coordination signal. The settlement layer is token-agnostic: any ERC-20 can serve as the bond and payment currency. This means:

- A stablecoin signals legal-system alignment.
- A DAO governance token signals community membership.
- A commodity-backed token signals value anchoring.
- Accepting a specific token *is* the seller's identity declaration—an accepted-token list is a brand.

Settlement velocity in a given token replaces traditional reputation metrics (star ratings, review scores). The on-chain record of resolved orders denominated in token τ is a public, tamper-proof signal of the seller's reliability within the τ economy. No platform needs to aggregate, curate, or attest this signal; it is a natural byproduct of protocol usage.

This matters institutionally because reputation infrastructure is one of the durable rents that platforms extract. A standing platform's credentialing function—certifying that a counterparty is reliable, safe, or qualified—is one of the structural reasons platforms persist even after their matching function is technically replicable. When reputation is a public byproduct of settlement rather than a proprietary platform asset, the platform's claim to that rent weakens.

6 Cultural Hegemony and the Visibility of Coordination

Gramsci [1971] observed that the most durable forms of power are those embedded so deeply in common sense that they become invisible. He called this *cultural hegemony*: the dominant arrangements rule not only through coercion but by making themselves appear natural, inevitable, and beyond question. The concept applies directly to contemporary coordination infrastructure.

The platform economy illustrates Gramscian hegemony precisely. The proposition that two strangers need a platform to transact safely feels like common sense. It is not. It is a contingent arrangement that arose because no economic primitive existed for safe direct exchange between strangers. In Gramscian terms, the platform has achieved hegemonic status: its necessity is no longer argued for—it is assumed.

Similarly, the proposition that production requires firms (the Coasean assumption examined in Theorem 3.1) and that cross-border trade requires corridors controlled by state or alliance actors (ports, railways, digital infrastructure) have both achieved the same hegemonic status. These are presented as structural necessities rather than contingent design choices.

The primitive makes the coordination function *visible* by separating it from the entities that currently perform it. The platform's coordination function (matching, enforcement, dispute resolution) becomes a public smart contract. The firm's coordination function (search, negotiation, enforcement) becomes three reducible costs. The corridor's coordination function (settlement, standards, access control) becomes a bilateral mechanism. In each case, the entity does not need to be *replaced*—it needs to be *seen* as a design choice rather than a fact of nature. Once the coordination function is visible and separable, the architecture that makes the entity structurally unnecessary follows.

This analysis explains a practical difficulty in communicating the implications of the primitive. Proposing to remove structures that most people do not know are there is qualitatively different from proposing a better version of the same structure. The challenge is not engineering. It is making existing infrastructure legible as architecture rather than gravity.

A scope caveat. The Gramscian framing identifies a barrier to recognition; it does not, by itself, predict which institutions will dissolve and which will persist. The threshold proposition (Theorem 3.1) and the firm-type distinction (Section 4) do that work. The hegemony argument explains why the work is needed: not because the institutional consequence is empirically subtle, but because the incumbent arrangement is presupposed in the language used to reason about alternatives.

7 Conclusion

The Coasean threshold is not a conjecture—it is observable. Contemporary coordination stacks often extract 15–30% of transaction value as coordination rent (Uber 25%, DoorDash 30%, Airbnb 14%, Amazon Marketplace 15%). Credit card networks layer 2–3% interchange plus double-digit annual interest on revolving balances. Financial markets intermediate trillions in notional value against a fraction of that in real economic exchange. Asymmetric bonding prices the cost of trust at $2\times$ the transaction value in temporarily locked (not spent) capital, with zero recurring extraction.

For any economic activity where the coordination rent exceeds the opportunity cost of locked capital, the standing firm—as Coase conceived it—is no longer the uniquely efficient unit of economic organization in that activity. The stronger consequence is the emergence of transaction-scoped institutions: temporary assemblies of directly bonded contributors that form around a process and dissolve at settlement. The settlement primitive does not abolish coordination; it changes the unit of coordination from the standing firm to the bonded process—in the regions where it is applicable.

The argument leaves several questions open. First, the empirical question: which industries and which transaction classes lie above the threshold today, and at what rate does the threshold shift as on-chain settlement matures? Second, the institutional question: how do the firm functions that the primitive does not address—capital aggregation, organizational knowledge, risk pooling, legal personhood—reconstitute around the primitive without re-creating the coordination firm by the back door? Third, the political question: who pays the transition costs of replacing standing intermediaries with bilateral mechanisms, and what coalitions form to slow or accelerate the shift? Each is empirical or institutional rather than mechanism-theoretic; the present paper has limited itself to identifying that the questions are now well-posed.

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